

5.7 Classwork: Calculus Practice — Markscheme

1. Let $f(x) = x^3 - 4x + 1$.

(a) Find $f'(x)$. [2]

$$f'(x) = 3x^2 - 4 \quad (\mathbf{A1})(\mathbf{A1})$$

(b) Find the gradient of the graph of f at $x = 2$. [2]

$$f'(2) = 3(4) - 4 \quad (\mathbf{M1})$$

$$= 8 \quad (\mathbf{A1})$$

(c) Find the equation of the tangent line at $x = 2$. [3]

$$f(2) = 8 - 8 + 1 = 1 \quad (\mathbf{A1})$$

$$y - 1 = 8(x - 2) \quad (\mathbf{M1})$$

$$y = 8x - 15 \quad (\mathbf{A1})$$

(d) Find the coordinates of the points where the gradient of f is zero. [2]

$$3x^2 - 4 = 0 \Rightarrow x = \pm \frac{2}{\sqrt{3}} = \pm \frac{2\sqrt{3}}{3} \quad (\mathbf{A1})$$

$$\left(-\frac{2\sqrt{3}}{3}, 1 + \frac{16\sqrt{3}}{9}\right) \text{ and } \left(\frac{2\sqrt{3}}{3}, 1 - \frac{16\sqrt{3}}{9}\right) \quad (\mathbf{A1})$$

2. Let $f(x) = 6x^2 - 2x$.

(a) Find $\int (6x^2 - 2x) dx$. [3]

$$\int (6x^2 - 2x) dx = 2x^3 - x^2 + c \quad (\mathbf{A1})(\mathbf{A1})(\mathbf{A1})$$

(b) Find the area of the shaded region. [3]

$$\int_1^2 (6x^2 - 2x) dx = [2x^3 - x^2]_1^2 \quad (\mathbf{M1})$$

$$= (16 - 4) - (2 - 1) \quad (\mathbf{A1})$$

$$= 12 - 1 = 11 \quad (\mathbf{A1})$$

(c) Find $g(x)$ given $g'(x) = 6x^2 - 2x$ and $g(1) = 7$. [4]

$$g(x) = 2x^3 - x^2 + c \tag{M1}$$

$$g(1) = 2 - 1 + c = 7 \tag{M1}$$

$$c = 6 \tag{A1}$$

$$g(x) = 2x^3 - x^2 + 6 \tag{A1}$$

3. Let $f(x) = (2x + 1)e^x$.

(a) Show that $f'(x) = (2x + 3)e^x$. [3]

$$\text{Product rule: } f'(x) = 2 \cdot e^x + (2x + 1) \cdot e^x \quad (\text{M1})$$

$$= e^x(2 + 2x + 1) \quad (\text{A1})$$

$$= (2x + 3)e^x \quad (\text{A1}) \quad (\text{AG})$$

(b) Find the equation of the tangent at $x = 0$. [3]

$$f(0) = (1)(1) = 1 \quad (\text{A1})$$

$$f'(0) = (3)(1) = 3 \quad (\text{A1})$$

$$y = 3x + 1 \quad (\text{A1})$$

(c) Find the coordinates of A , where the tangent meets the x -axis. [2]

$$0 = 3x + 1 \Rightarrow x = -\frac{1}{3} \quad (\text{M1})$$

$$A = \left(-\frac{1}{3}, 0\right) \quad (\text{A1})$$

(d) Find the x -coordinate of the local minimum. Justify. [4]

$$f'(x) = 0 \Rightarrow (2x + 3)e^x = 0 \quad (\text{M1})$$

$$e^x > 0 \text{ for all } x, \text{ so } 2x + 3 = 0$$

$$x = -\frac{3}{2} \quad (\text{A1})$$

$$f''(x) = (2x + 5)e^x \quad (\text{M1})$$

$$f''\left(-\frac{3}{2}\right) = 2 \cdot e^{-3/2} > 0 \quad (\text{R1})$$

Therefore it is a minimum.

4. $v(t) = t^2 - 5t + 4$, $s(0) = 2$.

(a) Write down the velocity when $t = 0$. [1]

$$v(0) = 0 - 0 + 4 = 4 \text{ m s}^{-1} \quad (\text{A1})$$

(b) Find the times when the particle is at rest. [3]

$$t^2 - 5t + 4 = 0 \quad (\text{M1})$$

$$(t - 1)(t - 4) = 0 \quad (\text{A1})$$

$$t = 1 \text{ and } t = 4 \quad (\text{A1})$$

(c) Find the acceleration when $t = 3$. [3]

$$a(t) = v'(t) = 2t - 5 \quad (\text{M1})(\text{A1})$$

$$\begin{aligned} a(3) &= 6 - 5 \\ &= 1 \text{ m s}^{-2} \end{aligned} \quad (\text{A1})$$

(d) Find $s(t)$. [4]

$$s(t) = \int v \, dt = \frac{t^3}{3} - \frac{5t^2}{2} + 4t + c \quad (\text{M1})(\text{A1})$$

$$s(0) = 2 \Rightarrow c = 2 \quad (\text{M1})$$

$$s(t) = \frac{t^3}{3} - \frac{5t^2}{2} + 4t + 2 \quad (\text{A1})$$

(e) Find the total distance in the first 4 seconds. [3]

v changes sign at $t = 1$ and $t = 4$ (M1)

$$\int_0^1 v \, dt = \left[\frac{t^3}{3} - \frac{5t^2}{2} + 4t \right]_0^1 = \frac{1}{3} - \frac{5}{2} + 4 = \frac{11}{6} \quad (\text{A1})$$

$$\int_1^4 v \, dt = \left[\frac{t^3}{3} - \frac{5t^2}{2} + 4t \right]_1^4 = \frac{-8}{3} - \frac{11}{6} = \frac{-9}{2} \quad (\text{A1})$$

$$\text{Total distance} = \frac{11}{6} + \frac{9}{2} = \frac{11}{6} + \frac{27}{6} = \frac{38}{6} = \frac{19}{3} \text{ m}$$